

Course Name: Data Structures using C++

Course Code: PG-C-AIML-415PT

Credits: 4

Course Outcomes:

The objective of the course is to enable students to

- **CO1:** Understand the fundamentals of programming in C++ including syntax, control structures, functions, arrays, and object-oriented programming concepts
- **CO2:** Design and implement modular programs using classes, objects, constructors, inheritance, and file handling techniques
- **CO3:** Analyse and implement fundamental data structures such as arrays, linked lists, stacks, queues, trees, and graphs using C++
- **CO4:** Apply algorithmic design techniques for searching, sorting, recursion, and basic problem-solving while evaluating algorithm efficiency using asymptotic analysis

Prerequisites:

Knowledge of C language, Programming concepts including algorithm building and logical problem solving

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Basics of C++ <i>Introduction to C++:</i> Structure of a C++ Program, Tokens, Keywords, Identifiers, Constants, Variables and data types, Operators and Expressions, Pointers, Input and Output Operations <i>Control Statements and Functions:</i> Conditional and Looping Constructs, Arrays and Strings, Declaration and Definition of a Functions, Parameter Passing, Recursion and Inline Functions	01	30 Hrs.
2.	Unit Title: Introduction to Object Oriented Programming <i>Concepts of Object-Oriented Programming:</i> Classes and Objects, Access Specifiers, Constructors and Destructors, Function Overloading, Operator Overloading, Friend Functions and Friend Classes <i>Inheritance:</i> Single, Multilevel, Hierarchical and Multiple Inheritance, Virtual Functions and Polymorphism	01	30 Hrs.

Unit No.	Particulars	Credits	Time
3.	Unit Title: Linear Data Structures <i>Arrays and Linked Lists:</i> Difference between Static and Dynamic Memory Allocation, Arrays (1D&2D) Implementation and their Operations, Implementation of Linked Lists (Singly, Doubly, Circular) and their applications <i>Stacks and Queues:</i> Implementation of Stacks using Array and Linked list, Applications of Stacks- Infix to Postfix, Postfix Evaluation & Matching of Parenthesis, Implementation of Queue (Linear and Circular) using Array and Linked List	01	30 Hrs.
4.	Unit Title: Non-Linear Data Structure and Algorithms <i>Trees:</i> Tree Terminologies, binary trees, Implementation of Binary Search Trees (Traversals and Searching), heap <i>Graphs:</i> Terminologies of Directed and Indirected Graphs, Representation using Adjacency Matrix and Adjacency List, Graph Traversal using BFS and DFS, Minimum Spanning Tree (Prim's and Kruskal Algorithms), Searching Algorithms (Linear & Binary Search), Sorting Algorithms (Bubble, Insertion, Selection)	01	30 Hrs.

References:

1. Bjarne Stroustrup, "The C++ Programming Language", Addison-Wesley
2. E. Balagurusamy, "Object Oriented Programming with C++", McGraw-Hill
3. Seymour Lipschutz, "Data Structures with C", Schaum's Outline Series
4. Ellis Horowitz, Sartaj Sahni, and Susan Anderson-Freed, "Fundamentals of Data Structures in C++", Universities Press
5. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, "Introduction to Algorithms", MIT Press

Accomplishments of the student after completing the course:

At the end of the course, the students will be able to

- Develop structured and object-oriented programs using C++
- Apply inheritance and polymorphism to design reusable software components
- Implement and analyze core data structures for efficient data organization
- Design and evaluate algorithms for searching, sorting, and problem solving
